PROJECT BLUEPRINT

Collective Procurement Platform

A managed B2B buying network that gives independent retailers the collective purchasing power to negotiate better wholesale prices.

CORE MODEL
Collective buying + managed procurement + platform-controlled fulfilment

THREE PILLARS
Collective Buying • Intelligent Procurement • Collective Negotiation

OPERATING CHOICE
Virtual distribution hub • Platform-managed delivery coordination • Centralised checkout

Prepared as an implementation blueprint for a fast, demo-ready web application. The MVP uses simulated payment and logistics events where real integrations would exceed the build window.

PRODUCT CONCEPT • BUSINESS MODEL • ORDER WORKFLOWS • EXCEPTION HANDLING • MVP BUILD PLAN

1. Executive summary

The product is a managed B2B procurement platform connecting small retailers with wholesalers. Instead of simply listing suppliers, the platform aggregates demand from multiple retailers, helps them reach a wholesaler's minimum order threshold, manages supplier selection and checkout, and coordinates delivery to each retailer.

The platform is the transaction coordinator. Retailers place orders through the platform; wholesalers receive consolidated purchase orders; the platform tracks each retailer's allocation, payment state, fulfilment and delivery.

Problem being solved

- Small retailers often buy in lower volumes and may not receive the same wholesale rates or bargaining power as large retail chains.
- Retailers may rely on a limited set of distributors and find it difficult to compare effective prices, minimum order quantities and delivery costs.
- Wholesalers want larger, predictable orders and a broader customer base without individually coordinating many small orders.

Proposed value

- **For retailers:** access to pooled volume discounts, transparent pricing, convenient ordering and delivery coordination.
- **For wholesalers:** aggregated demand, fewer fragmented order conversations and access to new retailer customers.
- **For the platform:** transaction commission and optional supplier or logistics-related revenue, subject to real-world unit economics.

Positioning statement

Give independent retailers the collective buying power of large retail chains. The platform makes wholesale procurement easier by pooling demand, coordinating supplier competition and managing the order journey end to end.

2. Product pillars and feature scope

The product should be organised around three clear pillars. Features should support these pillars rather than becoming a disconnected list of gimmicks.

Pillar 1 — Collective Buying

- Group orders: retailers request quantities of the same product and contribute to one shared order.
- Threshold meter: show required quantity, committed quantity, remaining quantity and deadline.
- Neighbourhood buying clubs: reusable groups of retailers in a locality or delivery zone.
- Group allocation: preserve each retailer's individual quantity, price, payment and delivery record.

Pillar 2 — Intelligent Procurement

- Rule-based recommendations using order history, saved products, stock levels and recent buying patterns.
- Supplier comparison based on effective landed price: product price + taxes/fees + delivery cost, not just headline price.
- Basic reorder reminders and frequently purchased items.
- Analytics: spend, estimated savings, purchase frequency and supplier reliability. Use historical statistics rather than promising predictive AI in the first version.

Pillar 3 — Collective Negotiation

- Reverse auction: retailers' aggregated demand is presented to eligible wholesalers, who submit bids.
- Smart negotiation rules: compare price, MOQ, availability, delivery promise and supplier reliability.
- Backup supplier shortlist: keep an alternative bid or supplier available before confirming a group.

Additional differentiators

Feature	MVP behaviour
Neighbourhood Buying Clubs	Retailers join locality-based groups and see pooled demand.
Emergency Inventory Exchange	Retailers list surplus stock for nearby retailers; platform records the transfer.
Retailer Power Index	A transparent score based on completed pooled purchases, savings and reliability—not an opaque credit score.
Supplier Reliability Score	Based on fulfilment, on-time delivery, cancellations and dispute outcomes.
Price statistics	Show historical observed prices and average savings from platform transactions; avoid forecasting claims initially.

3. Users, dashboards and transaction control

Role	Main responsibilities
Retailer	Browse products and group deals; create/join buying groups; commit quantity; pay; track allocation and delivery; raise disputes.
Wholesaler	Manage catalogue, stock, MOQ and pricing; respond to procurement requests; submit bids; accept purchase orders; update fulfilment.
Platform admin / operations	Verify suppliers; moderate products; oversee groups and auctions; approve supplier selection; coordinate virtual hub and delivery; handle disputes and refunds.

The platform should have role-based dashboards, not three separate products. Shared components and a common design system should keep development fast.

End-to-end procurement flow

Step	What happens
1. Discover	Retailer searches or reorders a product and sees eligible suppliers, MOQ, pricing slabs and delivery zones.
2. Join demand	Retailer enters quantity and joins an existing group or creates a new demand pool.
3. Aggregate	Platform combines eligible demand for the same SKU / pack size / delivery area and displays threshold progress.
4. Negotiate	Wholesalers submit bids; platform compares total landed cost, MOQ, availability and reliability.
5. Confirm	Platform selects a supplier under disclosed rules and locks the group after the commitment window.
6. Collect payment	Retailers pay their individual totals through central checkout. The demo simulates payment success/failure.
7. Purchase order	Platform sends one consolidated purchase order to the wholesaler with retailer-level allocation details.
8. Receive and sort	Virtual hub workflow records goods received, quantity checks, exceptions and retailer-wise parcels.
9. Deliver	Platform assigns delivery tasks to a delivery partner / route and tracks each retailer’s parcel.
10. Close	Retailer confirms delivery; receipt is available; disputes or refunds follow the resolution workflow.

Important operating principle

Retailers and wholesalers should not need to negotiate or settle orders independently. Contact details can remain masked until operationally necessary. The platform remains the system of record for order, payment status, allocation, delivery and dispute history.

4. Group-order lifecycle and protection rules

A group order needs explicit states. This makes the experience understandable to users and prevents ambiguous cancellations or payment expectations.

State	Meaning / allowed action
Open	Retailers can join, change quantity or withdraw under the displayed rules.
Near threshold	Platform sends progress alerts and invites matching retailers.
Threshold reached	Group becomes eligible for supplier confirmation and payment window.
Payment pending	Each retailer must pay by the stated deadline; unpaid allocations are not treated as confirmed.
Locked / confirmed	Quantities and price are committed; changes require admin or supplier-approved exception.
Supplier processing	Wholesaler prepares consolidated order.
At virtual hub	Goods are recorded, checked, sorted and allocated.
Out for delivery	Individual retailer shipments are assigned and tracked.
Delivered / disputed	Retailer confirms receipt or raises a supported issue.
Closed / refunded	Order is completed or cancelled and refund status is recorded.

Exception handling playbook

What can go wrong?	Recommended platform response
Retailer backs out before lock	Allow withdrawal according to the group rules; recalculate progress; notify the group; seek replacement demand.
Retailer backs out after lock	Do not silently remove the allocation. Flag for admin; attempt a replacement buyer; apply only a clearly disclosed cancellation policy. For demo, simulate an admin decision.
Threshold not met by deadline	Send reminders at configured milestones; extend once if feasible; try a lower-MOQ backup supplier or alternative group; otherwise cancel and show refund initiation.
Threshold falls after a withdrawal	Recalculate immediately; if below target, start a short recovery window and notify affected members.
Payment fails / expires	Mark that retailer’s allocation unpaid; notify them; offer retry; release allocation after deadline and attempt replacement.
Wholesaler withdraws / fails to confirm	Move to next eligible bid if price and terms remain valid; otherwise seek retailer approval for revised terms or cancel.
Wholesaler changes price	Reject unilateral price changes after bid acceptance. Any revised price requires explicit retailer consent before payment/lock.
Stock shortfall / partial fulfilment	Record received quantity; offer partial delivery, replacement or refund for missing units with retailer consent.
Delivery delay	Update revised ETA; notify affected retailers; allow support escalation.
Damaged / missing goods	Collect item-level issue, photos and delivery proof; open dispute; admin records resolution and refund/replacement.
Retailer refuses delivery	Record reason and evidence; route to support; apply disclosed return/cancellation rules.

5. Threshold recovery, notifications and safeguards

Threshold recovery should be automatic and predictable. The platform should not keep a group open indefinitely or quietly change its price.

Suggested threshold recovery sequence

Trigger	Automated action
Group created	Show MOQ, current quantity, deadline, price slabs and cancellation terms.
50% reached	Notify members that the group is progressing; recommend it to relevant nearby retailers.
80% reached	Send near-threshold alert; show exact remaining units and time left.
24 hours remaining	Send deadline reminder; prompt retailers to share the deal with eligible peers.
Deadline reached, below MOQ	Try one configurable extension or offer backup supplier / alternate MOQ.
Recovery fails	Cancel group, mark payments for refund processing and explain the reason to every member.

Notification centre

- Group joined; quantity changed; group is 80% complete; deadline approaching; threshold reached.
- Payment due; payment failed; order locked; supplier confirmed; supplier changed / backup supplier selected.
- Goods received at virtual hub; quantity mismatch; parcel dispatched; delivery delayed; out for delivery; delivered.
- Dispute opened; more information required; resolution issued; refund initiated / completed.

Commitment and fairness rules

- Display cancellation rules before the retailer joins and again before payment.
- Use a clear commitment lock after threshold, supplier confirmation and payment deadline—not merely when the quantity meter reaches 100%.
- Never charge a cancellation penalty that was not disclosed and accepted. For a demo, use admin-reviewed exceptions rather than complex automated penalties.
- Keep a timestamped event log for quantity changes, bids, price approvals, payments, delivery scans and dispute decisions.
- Protect individual retailer identity and purchase value. Show group totals and progress to other members unless the retailer opts in to sharing more.

6. Virtual hub, delivery and payment model

Virtual distribution hub

The virtual hub is a software workflow, not a physical warehouse. It represents the point where the consolidated wholesaler shipment is considered received, checked, split into retailer allocations and handed to delivery partners. In a real operation, a physical cross-dock or partner facility may be needed; the demo can simulate hub scans and status changes.

Hub stage	System action
Expected	Show supplier dispatch and expected arrival.
Received	Admin records shipment arrival and received quantity.
Quality / quantity check	Record shortage, damage or mismatch against purchase order.
Sorted	Generate retailer-wise pick lists and parcel IDs.
Ready for dispatch	Assign delivery partner, route / area and estimated delivery window.
Dispatched	Create tracking events for each retailer parcel.

Platform-managed delivery

- For the MVP, platform operations assign a simulated delivery partner and status updates.
- In a real rollout, begin with third-party local delivery partners or wholesaler transport under platform tracking; do not build a fleet immediately.
- Use delivery zones and route batching to combine nearby retailer drops.
- Track delivery charge separately and show it before payment. Avoid hiding logistics costs inside product prices.

Centralised checkout and receipts

- Retailer pays the platform checkout for their individual allocation; the system records payment status against the group order.
- Demo payment methods: UPI, card and COD as selectable simulated options. Never collect real card / UPI credentials in a demo.
- Generate an order receipt with order ID, retailer, supplier, SKU, quantity, unit price, discount, tax/fees if configured, delivery charge, total, payment method and status.
- Keep receipt separate from a tax invoice. A real GST invoice must be issued by the legally appropriate seller and contain the required tax details.

Real-world payment caution

For a real launch, use a payment provider that supports the intended marketplace / split-settlement flow and obtain legal and tax advice on who is the seller of record, who collects funds, refunds, GST invoicing and settlement timing. Do not describe an internal database status as escrow or hold customer funds yourself without an appropriate regulated arrangement.

7. Revenue model, trust and data

Revenue options

Revenue stream	How it works	MVP treatment
Transaction commission	Platform charges a disclosed fee on completed orders, preferably calculated on a clear order basis.	Calculate and display simulated platform revenue.
Logistics coordination margin	Platform charges a delivery / coordination fee and pays delivery partner according to agreed terms.	Show delivery charge and simulated cost / margin.
Wholesaler subscription	Optional paid supplier tools or enhanced catalogue / analytics visibility.	Show as a future plan, not required for MVP.
Promoted supplier placement	Clearly labelled sponsored visibility, subject to relevance and trust rules.	Future scope; do not let paid placement override bid quality.

About profit-sharing

The original example was: if a retailer earns ■100 additional profit, the platform keeps ■20 and the retailer keeps ■80. This is difficult to verify because the platform may not know the retailer's resale price, operating costs or actual realised margin. A transparent transaction commission or share of verified procurement savings is easier to measure. If savings-sharing is used, define the baseline price, calculation period and exclusions clearly.

Trust and supplier governance

- Supplier verification: business identity, GST details where applicable, bank / payout verification and operating area.
- Supplier performance: bid acceptance, fill rate, on-time delivery, cancellation rate and dispute rate.
- Product integrity: SKU, pack size, brand, quantity unit, expiry / batch information where relevant and substitution rules.
- Retailer safeguards: transparent final landed price, clear terms, delivery evidence and accessible support.

Core data entities

Entity	Key information
User	Role, contact, business profile, verification state, service area.
Product / SKU	Name, brand, pack size, unit, category, images, active status.
Supplier offer	Supplier, SKU, price slabs, MOQ, stock, delivery terms, validity.
Group order	SKU, zone, threshold, committed quantity, deadline, status, pricing rule.
Group member allocation	Retailer, quantity, unit price, payment state, cancellation state.
Bid	Supplier, offered price, available quantity, lead time, validity, status.
Purchase order	Selected supplier, consolidated quantity, accepted price, retailer allocations.
Payment / receipt	Amount, method, simulated / provider reference, status, receipt details.
Hub / delivery event	Parcel, status, timestamp, operator, proof / notes.
Dispute / refund	Reason, evidence, decision, amount, status, audit trail.

8. Demo-ready MVP architecture

Recommended stack

Layer	Choice	Purpose
Frontend	React + Vite	Role-based dashboards and responsive application.
UI	Tailwind CSS + shadcn/ui + Lucide	Consistent components and fast styling.
Backend	Node.js + Express	REST APIs, role checks and business workflows.
Database	MongoDB	Users, products, groups, bids, orders and events.
Authentication	JWT + role-based access	Retailer, wholesaler and admin access.
Charts	Recharts	Savings, order progress and operations analytics.
PDF	jsPDF	Demo receipts and downloadable summaries.
Realtime	Polling first; Socket.IO optional	Refresh threshold and auction progress.

Build the smallest believable end-to-end journey

- Seed sample retailers, wholesalers, products, supplier offers and locality-based groups.
- Make one full group order work from join → threshold → bids → supplier selection → simulated payment → purchase order → virtual hub → delivery → receipt.
- Use simple deterministic rules for recommendations and supplier ranking. Label simulated data honestly.
- Build admin controls for exceptional states so the demo can show backout, threshold failure, supplier withdrawal, shortage, delay and refund scenarios.

Must-have screens

Screen	Key content
Public / onboarding	Product explanation, sign in, role selection and demo account access.
Retailer dashboard	Active group deals, savings, reorder suggestions, orders and notifications.
Marketplace / group detail	Product, MOQ, price tiers, progress meter, deadline, terms and join action.
Negotiation / auction	Aggregated demand, supplier bids, landed-cost comparison and selected supplier.
Wholesaler dashboard	Catalogue, MOQ / price slabs, bid inbox, accepted purchase orders and fulfilment status.
Admin operations	Supplier verification, groups, bids, payment states, virtual hub, delivery, disputes and metrics.
Order detail	Retailer allocation, receipt, payment state, tracking timeline and support action.

9. 10–14 hour implementation plan

This is an aggressive single-developer demo target. It assumes familiarity with React, Express and MongoDB, a prepared UI direction, and seeded data. Prioritise one complete journey over many half-working features.

Time	Build target	Definition of done
0:00–1:00	Setup + schema	Project runs; models and seeded demo data exist.
1:00–2:00	Auth + roles	Retailer, wholesaler and admin demo access works.
2:00–4:00	Catalogue + group buying	Retailer can view products, join group and see threshold update.
4:00–5:30	Bids + supplier selection	Wholesaler can submit bid; admin can select supplier.
5:30–7:00	Order + payment simulation	Retailer can confirm allocation and simulate payment; receipt generated.
7:00–8:30	Virtual hub + delivery	Admin can record received/sorted/dispatched/delivered states.
8:30–10:00	Exceptions + notifications	Backout, threshold miss, supplier failure and dispute paths can be demonstrated.
10:00–12:00	UI polish + testing	Responsive layout, empty/loading/error states and complete demo flow.
12:00–14:00	Buffer / optional features	Buying clubs, inventory exchange, Power Index and charts if core flow is stable.

If time is running out, cut in this order

- First defer: real-time sockets, advanced AI, live maps, complex route optimisation and external integrations.
- Then simplify: reverse auction to admin-managed supplier bids; inventory exchange to a basic listing flow; retailer Power Index to a simple formula.
- Do not cut: group threshold logic, individual allocations, order status, payment simulation, receipt, and admin exception controls.

Suggested demo story

A retailer needs 30 units of a product, but the wholesaler’s MOQ is 300. Other retailers join the same group until the target is reached. Two wholesalers submit bids. The platform selects the best eligible offer using landed cost and reliability. Each retailer pays for their own allocation. The wholesaler receives one consolidated purchase order. The platform records goods at its virtual hub, splits the order into retailer parcels, assigns delivery and generates receipts. Finally, demonstrate one exception—such as a retailer backing out before lock or a threshold miss—and show the recovery workflow.

10. Assumptions, open decisions and risks

Current product decisions

- Managed marketplace: transactions and order records stay within the platform.
- Virtual hub: software-coordinated receiving, checking, sorting and dispatch; no physical warehouse in the demo.
- Platform-managed delivery: simulated dispatch assignment for demo; real operations can use delivery partners.
- Centralised checkout: simulated payment states for demo; no real money is collected.
- Initial product focus should be grocery / FMCG for a coherent demo dataset, with category-independent schema where practical.

Decisions to confirm before a real pilot

Decision	Why it matters
Who is the seller of record?	Determines invoice, tax, consumer / commercial obligations and payment flow.
Who owns goods at each stage?	Determines loss, damage, insurance and liability.
Who bears delivery cost and failed-delivery risk?	Affects unit economics and retailer terms.
What is the cancellation policy after lock?	Must balance group reliability with fair retailer protection.
How are refunds and supplier settlements handled?	Requires payment-provider, legal and accounting design.
What categories and geographies launch first?	Affects SKU standards, supplier onboarding, logistics density and operations.

Key risks to validate

- Retailer density: group buying only works when enough retailers in a delivery area need compatible SKUs at similar times.
- Supplier reliability: a low bid is not valuable if the wholesaler cannot fulfil the order.
- Logistics economics: sorting and last-mile delivery can erase wholesale savings. Show total landed cost before checkout.
- Working capital and settlement: payment collection, refunds and supplier payouts need a compliant, cash-flow-safe structure.
- Data quality: product pack sizes and units must match exactly before combining demand.

Success metrics for the demo

- Groups reaching MOQ; average time to reach threshold; average pooled quantity.
- Price difference versus baseline offer; estimated retailer savings; completed orders.
- Supplier fill rate; on-time delivery rate; cancellations; dispute rate.
- Platform revenue per completed order and delivery cost per drop.

Final product direction: Build a managed collective procurement network—not a supplier directory. The MVP should prove that pooled demand can unlock better terms and that the platform can coordinate the order reliably from retailer commitment to final delivery.

This document is a product and demo implementation blueprint, not legal, tax, payment or logistics advice. Real-world launch arrangements should be validated with qualified providers and advisers.